

GE Aerospace Stock Price Predictor

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Introduction

This report details a Recurrent Neural Network (rNN) developed to predict the closing price of GE Aerospace (NYSE: GE) on March 13th using 29 days of data. This model will then be applied to a different company within the same industry (Lockheed Martin, NYSE: LMT) and a company of a different industry (Exxon Mobile, NYSE: XOM). Performance metrics for all companies will be recorded to evaluate accuracy and reliability.

Methodology Dataset and Preprocessing

The dataset consists of historical daily stock prices for GE, LMT and XOM, including attributes such as Date, Open, Close, High, Low, and Volume. The primary target variable is the Closing Price. Before feeding data into the RNN, closing prices were normalized and transformed into overlapping sequences of 25 observations; any amount higher than this threshold resulted in model overfitting. 80% of the data was used for training, and 20% was held out for testing. The model predictions were finally converted to the original dollar-denominated scale for easier interpretation.

Model Architecture

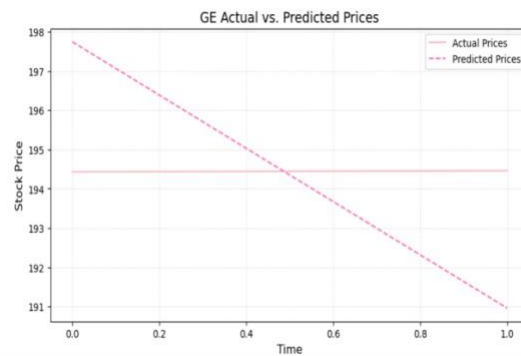
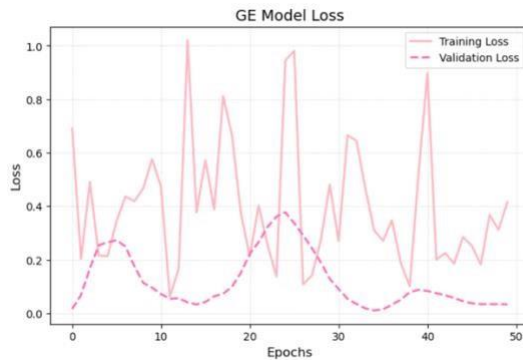
A Simple Recurrent Neural Network (RNN) was selected for this task due to its ability to handle sequential data. 50 recurrent neurons were chosen because higher values led to overfitting. A second RNN layer to further process sequence patterns. Dropout, set at 0.4, aids in preventing overfitting tendencies in the model. Epochs were set at 50 and batch size were at 64; any combination that deviates from these thresholds affected performance metrics (particularly, an increase in Root Mean Squared Error and Mean Absolute Error).

Closing Price Forecast Comparison

Company	Predicted	Actual	Error
GE Aerospace	\$190.96	\$192.49	0.79%
Lockheed Martin	\$455.21	\$467.87	2.71%
ExxonMobil	\$107.01	\$108.65	1.51%

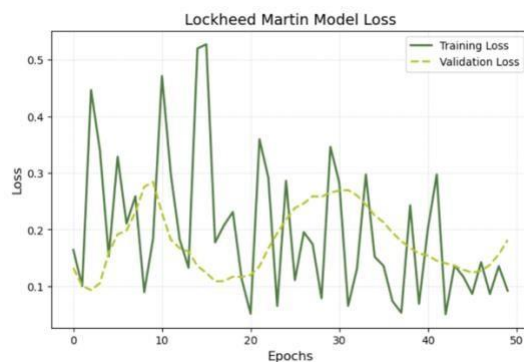
Absolute percentage differences, rounded to two decimal places. These are individual forecast comparisons, not the model's reported test-set MAE or RMSE

Assessment: GE Aerospace



The RNN produced a closing price forecast of \$190.96, compared with an actual closing price of \$192.49. The model's evaluation RMSE and MAE were \$3.41. The training-loss curve shows large fluctuations across epochs, while the validation loss generally peaks twice before stabilizing at a relatively low level. However, the prediction shows a noticeable discrepancy between the observed and predicted prices.

Assessment: Lockheed Martin



The RNN produced a reported closing-price forecast of \$455.21, compared with an actual closing price of \$467.87. The Lockheed Martin training-loss curve behaves similarly to the GE Aerospace with fluctuations in loss, but the MSE and MAE are significantly higher, at \$34.30 and \$24.93, respectively. While prediction prices remained relatively flat and downward, the actual prices increased significantly. This discrepancy could be due to the model's inability to capture the upward momentum in LM's stock, despite the similarities in the initial loss pattern with GE, suggesting that the RNN architecture might not be suitable for Lockheed Martin's stock behavior.

Assessment: ExxonMobil



The RNN produced a reported closing-price forecast of \$107.01, compared with an actual closing price of \$108.65. These observations suggest that the model captured some directional movement in the available data but did not fully reproduce the observed closing prices. The results provide preliminary evidence of pattern recognition within this particular time series, rather than establishing that energy-sector stocks are inherently more predictable.

Final Thoughts & Future Improvements

This study investigated the application of a stacked Simple Recurrent Neural Network to closing-price forecasting for GE Aerospace, Lockheed Martin, and ExxonMobil. The primary objective was to examine the performance of a common neural network architecture across financial time series from different industries. The model produced reported closing-price forecasts of \$190.96 for GE Aerospace, \$455.21 for Lockheed Martin, and \$107.01 for ExxonMobil. When compared with the supplied actual closing prices, the corresponding absolute percentage differences were approximately 0.79%, 2.71%, and 1.51%, respectively. The evaluation results also revealed differences in the model's ability to reproduce observed price trajectories. In particular, the Lockheed Martin predictions did not adequately reflect the upward movement displayed in the evaluation data, while the ExxonMobil predictions followed a similar upward direction but remained below the observed prices. These findings illustrate that applying the same neural network architecture to different financial time series can produce substantially different prediction patterns and errors.

However, several methodological limitations restrict the conclusions that can be drawn from this experiment. These include the limited historical observations, the use of closing prices as the sole predictive feature, preprocessing leakage, and the absence of an independent test set. The results should therefore be interpreted as an exploratory application of recurrent neural networks rather than evidence of a validated financial forecasting system.

Future development could investigate additional predictive features, including trading volume and macroeconomic indicators, as well as more complex recurrent architectures such as Long Short-Term Memory (LSTM) networks.